CS 135 — L08: List Abbreviations & Big-O

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List Abbreviations

`(list 1 2 3)` is shorthand for `(cons 1 (cons 2 (cons 3 empty)))`.

Counting Steps → Big-O

Measure time by the **number of substitutions**. For linear list recursion, steps ∝ list length `n`.

Linear example:

```
(define (len lst)
  (cond [(empty? lst) 0]
        [else (add1 (len (rest lst)))]))
```

Note: Built-ins like `length`, `append`, and `reverse` should be treated as **linear** in the size of their input lists for complexity reasoning.

Spotting Quadratic Work

```
(define (rev-bad xs)              ; quadratic: uses append each step
  (cond [(empty? xs) empty]
        [else (append (rev-bad (rest xs)) (list (first xs)))]))
```

Exercises

1. Classify the complexity of `map`, `filter`, `increasing?`, `insert`.
2. Rewrite `rev-bad` using an accumulator to get linear time.